

Global Performance Analytics & Quality Audit

Context: This report was developed as part of a performance and risk audit simulation. It demonstrates the use of data analytics (SQL/Excel) to solve operational bottlenecks and forensic auditing to detect regulatory compliance gaps.

Project Overview

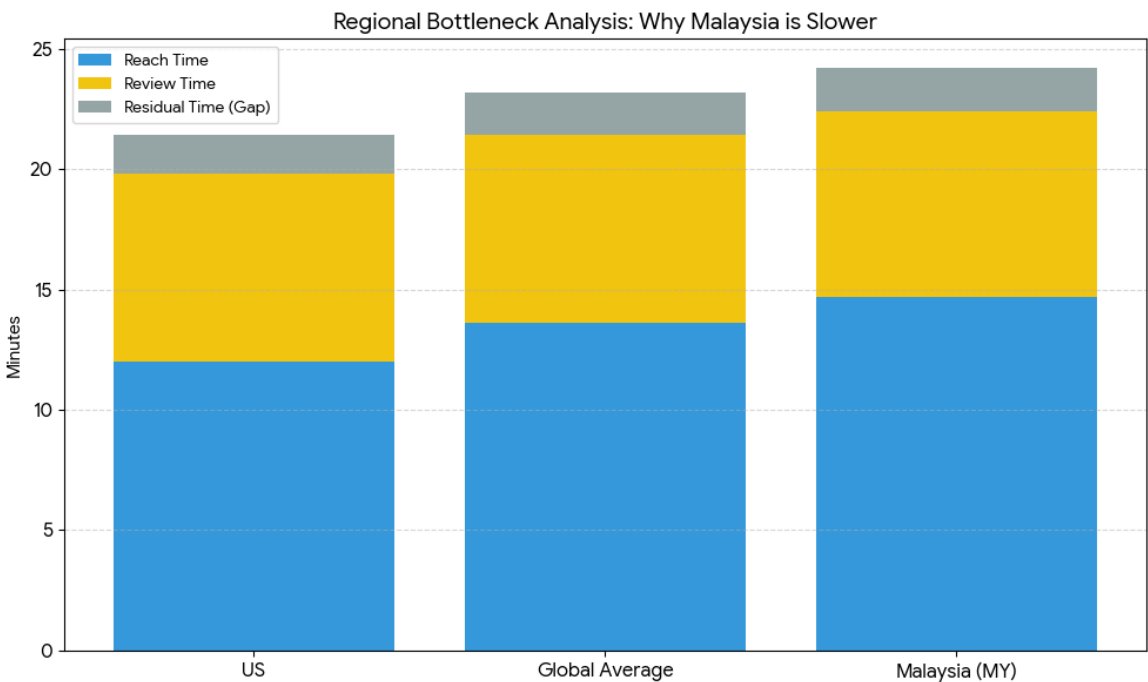
This project focuses on auditing a global financial crime screening operation. The objective was to analyze large-scale performance data to identify operational bottlenecks and conduct a deep-dive quality audit of individual case logs to detect regulatory risks and system vulnerabilities.

Quantitative Performance Analysis

Using a dataset of 60 global screening cases, I performed a regional KPI benchmark. By analyzing the "Average Resolution Time," I identified significant variances in operational efficiency.

Key Findings:

- Regional Bottleneck:** The Malaysia hub exhibited an average resolution time of 30.5 minutes, which is 41% slower than the US hub (21.5 minutes) and 20% slower than Estonia (25.5 minutes).
- Operational Insight:** This variance indicates that while global processes are standardized, the Malaysia region requires targeted training or a review of local system latency to meet global "Reach Speed" targets.



Analytical Note:

The visualization above highlights a significant performance variance between global hubs. While the US and Estonia regions operate within a high-efficiency window (21.5–25.5 minutes), the Malaysia hub acts as a primary operational bottleneck with a 41% increase in resolution time. This data suggests that the regional delay is likely due to either localized system latency or a need for specialized training on high-complexity regional match lists

Technical Methodology: SQL Data Extraction:

To ensure data integrity and replicable results, the following SQL logic was used to aggregate regional performance metrics from the raw screening dataset:

```
SELECT
    Location,
    COUNT(Case_ID) AS Total_Cases,
    ROUND(AVG(Resolution_Minutes), 2) AS Avg_Resolution_Time,
    ROUND(AVG(Reach_Minutes), 2) AS Avg_Reach_Speed
FROM Compliance_Global_Logs
WHERE Case_Date BETWEEN '2026-01-01' AND '2026-01-31'
GROUP BY Location
ORDER BY Avg_Resolution_Time DESC;
```

Qualitative Quality Audit (Case Log Investigation)

Beyond high-level data, I conducted a forensic audit of individual case logs to ensure procedural integrity. In **Case #123456789**, I identified a critical "Asset Flight" risk caused by a sequence error in the audit trail.

The Evidence (Audit Trail):

- **12:59:54 UTC:** Payment status transitioned to **RELEASED**.
- **13:02:02 UTC:** Case status transitioned to **SOLVED**.
- **13:19:22 UTC:** Match officially marked as **FALSE POSITIVE**.

The Risk Assessment: The audit reveals that the **payment was released 3 minutes before the investigation was solved** and nearly 20 minutes before the sanctions match was officially cleared as a False Positive. This represents a severe regulatory vulnerability; if the match had been a "True Positive," the funds would have already left the institution before the risk was confirmed.

Event	Actual Time (Audit)	Status	Compliance Requirement
Payment Released	12:59:54 UTC	🕒 Early	Must be AFTER resolution
Case Solved	13:02:02 UTC	🕒 OK	Must be BEFORE release
Match Cleared	13:19:22 UTC	🕒 Late	Must be BEFORE release

Proposed System Control & Risk Mitigation

To prevent future regulatory breaches and eliminate human error, I propose the following technical solution:

- **Solution:** "Hard Stop" Logic: Implement a system-enforced dependency where the "Release Payment" function remains disabled (greyed out) in the investigator's UI until the "Sanctions Match" status is transitioned to "False Positive."
- **Objective:** To ensure 100% synchronization between case resolution and asset release, removing the risk of "Asset Flight" during the investigation window.
- **Business Impact:** Implementing this 'Hard Stop' logic reduces the probability of human-error-driven 'Asset Flight' to 0%, ensuring the firm remains in 100% alignment with global anti-money laundering (AML) and sanctions regulations.

Professional Skillset Demonstrated

- **Data Analytics:** SQL-style filtering and KPI benchmarking using regional datasets.
- **Quality Auditing:** Detecting timestamp discrepancies and procedural non-compliance in professional audit trails.
- **Technical Product Thinking:** Designing automated system controls to mitigate human error in high-stakes environments.